

casestudy1

March 18, 2026

```
[1]: !pip install tensorflow
```

```
Requirement already satisfied: tensorflow in c:\users\mandi\anaconda3\lib\site-  
packages (2.21.0)  
Requirement already satisfied: absl-py>=1.0.0 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (2.4.0)  
Requirement already satisfied: astunparse>=1.6.0 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (1.6.3)  
Requirement already satisfied: flatbuffers>=25.9.23 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (25.12.19)  
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (0.7.0)  
Requirement already satisfied: google_pasta>=0.1.1 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (0.2.0)  
Requirement already satisfied: libclang>=13.0.0 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (18.1.1)  
Requirement already satisfied: opt_einsum>=2.3.2 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (3.4.0)  
Requirement already satisfied: packaging in c:\users\mandi\anaconda3\lib\site-  
packages (from tensorflow) (25.0)  
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (7.34.0)  
Requirement already satisfied: requests<3,>=2.21.0 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (2.32.5)  
Requirement already satisfied: setuptools in c:\users\mandi\anaconda3\lib\site-  
packages (from tensorflow) (80.9.0)  
Requirement already satisfied: six>=1.12.0 in c:\users\mandi\anaconda3\lib\site-  
packages (from tensorflow) (1.17.0)  
Requirement already satisfied: termcolor>=1.1.0 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (3.3.0)  
Requirement already satisfied: typing_extensions>=3.6.6 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (4.15.0)  
Requirement already satisfied: wrapt>=1.11.0 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (1.17.0)  
Requirement already satisfied: grpcio<2.0,>=1.24.3 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (1.78.0)  
Requirement already satisfied: keras>=3.12.0 in  
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (3.13.2)
```

Requirement already satisfied: numpy>=1.26.0 in
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (2.3.5)
Requirement already satisfied: h5py<3.15.0,>=3.11.0 in
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (3.14.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in
c:\users\mandi\anaconda3\lib\site-packages (from tensorflow) (0.5.4)
Requirement already satisfied: charset_normalizer<4,>=2 in
c:\users\mandi\anaconda3\lib\site-packages (from
requests<3,>=2.21.0->tensorflow) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in
c:\users\mandi\anaconda3\lib\site-packages (from
requests<3,>=2.21.0->tensorflow) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
c:\users\mandi\anaconda3\lib\site-packages (from
requests<3,>=2.21.0->tensorflow) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
c:\users\mandi\anaconda3\lib\site-packages (from
requests<3,>=2.21.0->tensorflow) (2026.1.4)
Requirement already satisfied: wheel<1.0,>=0.23.0 in
c:\users\mandi\anaconda3\lib\site-packages (from astunparse>=1.6.0->tensorflow)
(0.45.1)
Requirement already satisfied: rich in c:\users\mandi\anaconda3\lib\site-
packages (from keras>=3.12.0->tensorflow) (14.2.0)
Requirement already satisfied: namex in c:\users\mandi\anaconda3\lib\site-
packages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in c:\users\mandi\anaconda3\lib\site-
packages (from keras>=3.12.0->tensorflow) (0.19.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in
c:\users\mandi\anaconda3\lib\site-packages (from
rich->keras>=3.12.0->tensorflow) (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
c:\users\mandi\anaconda3\lib\site-packages (from
rich->keras>=3.12.0->tensorflow) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in c:\users\mandi\anaconda3\lib\site-
packages (from markdown-it-py>=2.2.0->rich->keras>=3.12.0->tensorflow) (0.1.2)

```
[2]: import pandas as pd
import numpy as np
```

```
[3]: train_df = pd.read_csv("Documents/train.csv")
test_df = pd.read_csv("Documents/test.csv")

print("Train shape:", train_df.shape)
print("Test shape:", test_df.shape)
```

Train shape: (42000, 785)
Test shape: (28000, 784)

```
[4]: X_train = train_df.drop("label", axis=1)
      y_train = train_df["label"]

      X_test = test_df

      print("X_train:", X_train.shape)
      print("y_train:", y_train.shape)
      print("X_test:", X_test.shape)
```

```
X_train: (42000, 784)
y_train: (42000,)
X_test: (28000, 784)
```

```
[5]: X_train = X_train / 255.0
      X_test = X_test / 255.0
```

```
[6]: import numpy as np

      X_train = np.array(X_train)
      y_train = np.array(y_train)
      X_test = np.array(X_test)
```

```
[7]: from sklearn.model_selection import train_test_split

      X_train, X_val, y_train, y_val = train_test_split(
          X_train, y_train,
          test_size=0.1,
          random_state=42
      )

      print("Training:", X_train.shape)
      print("Validation:", X_val.shape)
```

```
Training: (37800, 784)
Validation: (4200, 784)
```

```
[8]: import tensorflow as tf
      from tensorflow import keras

      model = keras.Sequential([
          keras.Input(shape=(784,)),
          keras.layers.Dense(256, activation='relu'),
          keras.layers.Dropout(0.3),
          keras.layers.Dense(128, activation='relu'),
          keras.layers.Dense(10, activation='softmax')
      ])
```

```

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

history = model.fit(
    X_train, y_train,
    epochs=15,
    batch_size=64,
    validation_data=(X_val, y_val)
)

```

WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

Epoch 1/15

591/591 9s 12ms/step -

accuracy: 0.9001 - loss: 0.3332 - val_accuracy: 0.9502 - val_loss: 0.1576

Epoch 2/15

591/591 7s 11ms/step -

accuracy: 0.9561 - loss: 0.1437 - val_accuracy: 0.9633 - val_loss: 0.1169

Epoch 3/15

591/591 10s 11ms/step -

accuracy: 0.9652 - loss: 0.1093 - val_accuracy: 0.9652 - val_loss: 0.1096

Epoch 4/15

591/591 6s 11ms/step -

accuracy: 0.9724 - loss: 0.0886 - val_accuracy: 0.9724 - val_loss: 0.0888

Epoch 5/15

591/591 5s 9ms/step -

accuracy: 0.9773 - loss: 0.0724 - val_accuracy: 0.9719 - val_loss: 0.0899

Epoch 6/15

591/591 7s 12ms/step -

accuracy: 0.9796 - loss: 0.0619 - val_accuracy: 0.9762 - val_loss: 0.0829

Epoch 7/15

591/591 6s 10ms/step -

accuracy: 0.9815 - loss: 0.0564 - val_accuracy: 0.9764 - val_loss: 0.0806

Epoch 8/15

591/591 9s 9ms/step -

accuracy: 0.9821 - loss: 0.0548 - val_accuracy: 0.9729 - val_loss: 0.0869

Epoch 9/15

591/591 7s 12ms/step -

accuracy: 0.9849 - loss: 0.0449 - val_accuracy: 0.9771 - val_loss: 0.0832

Epoch 10/15

591/591 7s 11ms/step -

accuracy: 0.9853 - loss: 0.0431 - val_accuracy: 0.9760 - val_loss: 0.0838

Epoch 11/15

```

591/591          11s 12ms/step -
accuracy: 0.9872 - loss: 0.0384 - val_accuracy: 0.9767 - val_loss: 0.0916
Epoch 12/15
591/591          8s 9ms/step -
accuracy: 0.9874 - loss: 0.0389 - val_accuracy: 0.9783 - val_loss: 0.0831
Epoch 13/15
591/591          8s 13ms/step -
accuracy: 0.9890 - loss: 0.0318 - val_accuracy: 0.9786 - val_loss: 0.0749
Epoch 14/15
591/591          8s 9ms/step -
accuracy: 0.9885 - loss: 0.0340 - val_accuracy: 0.9783 - val_loss: 0.0848
Epoch 15/15
591/591          11s 11ms/step -
accuracy: 0.9901 - loss: 0.0292 - val_accuracy: 0.9786 - val_loss: 0.0847

```

```

[9]: val_loss, val_acc = model.evaluate(X_val, y_val)
    print("Validation Accuracy:", val_acc)

```

```

132/132          1s 5ms/step -
accuracy: 0.9786 - loss: 0.0847
Validation Accuracy: 0.9785714149475098

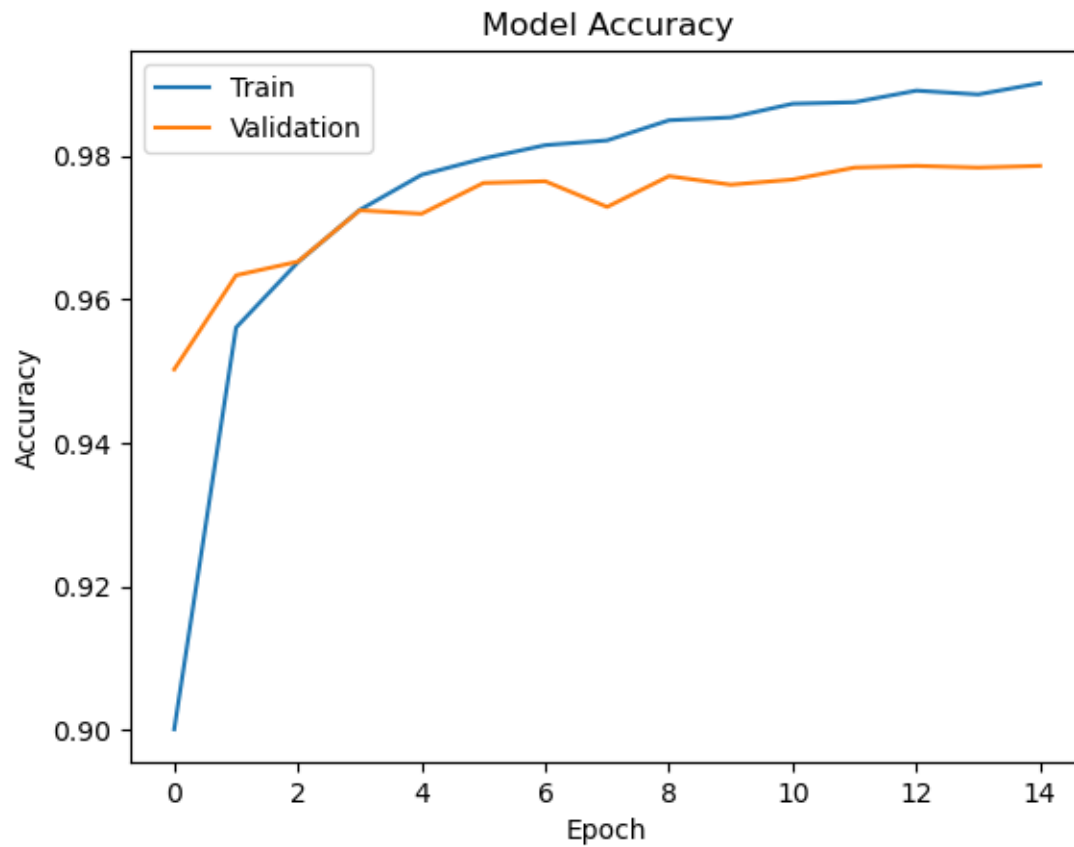
```

```

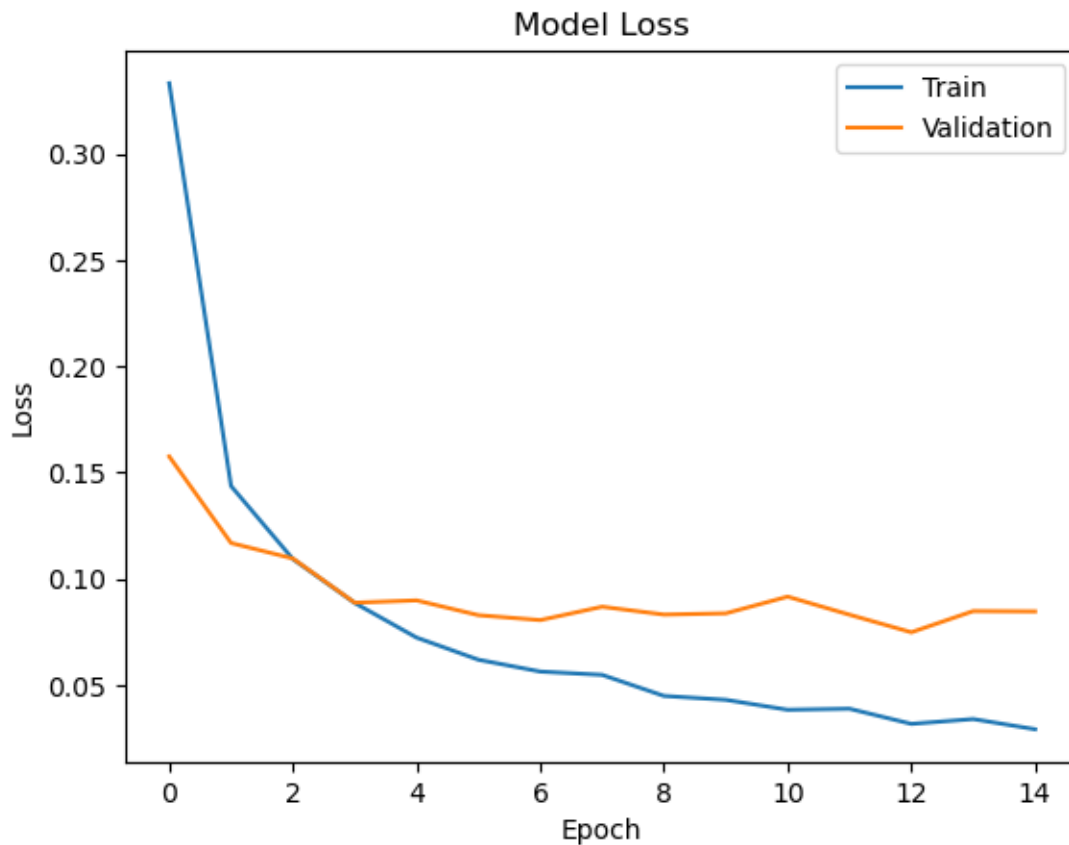
[10]: import matplotlib.pyplot as plt

plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title('Model Accuracy')
plt.ylabel('Accuracy')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'])
plt.show()

```



```
[11]: plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title('Model Loss')
plt.ylabel('Loss')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'])
plt.show()
```



```
[12]: predictions = model.predict(X_test)
      predicted_labels = predictions.argmax(axis=1)
```

875/875 2s 3ms/step

```
[13]: submission = pd.DataFrame({
      "ImageId": range(1, len(predicted_labels) + 1),
      "Label": predicted_labels
    })

      submission.to_csv("submission.csv", index=False)
```